

DishDetect: What's on my plate? – Co-Designing A Mobile App for Clockwise Food Description

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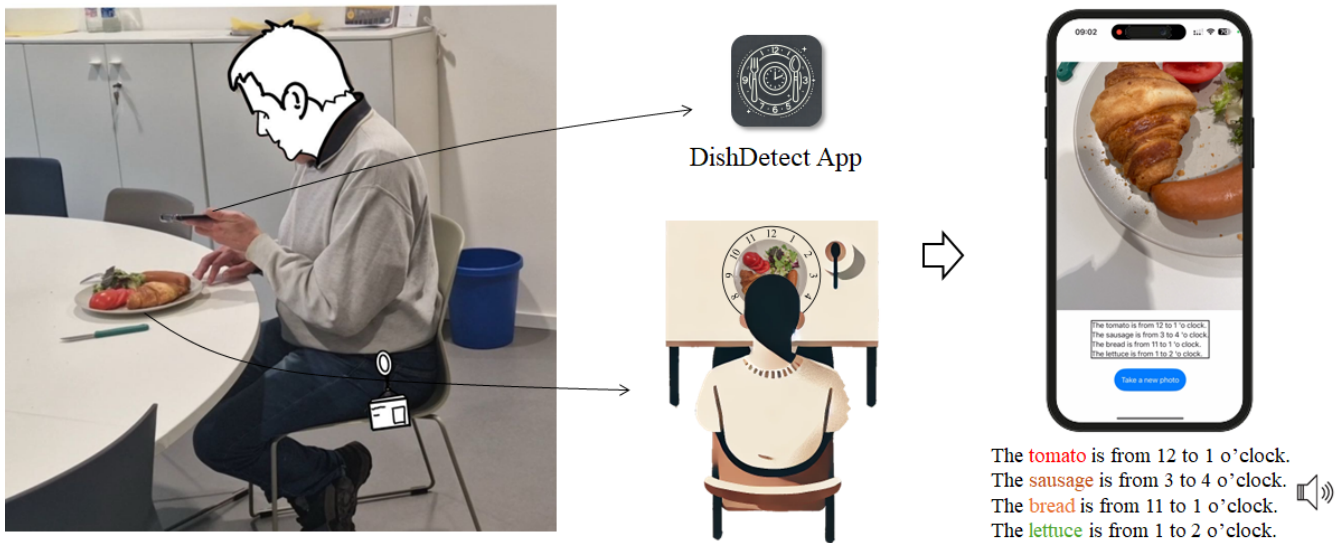


Figure 1: DishDetect for clockwise food description.

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Abstract

This work presents the design and development of a mobile application with and for people with blindness that helps to find out where and what is on a plate. The semantic segmentation algorithm, the proposed system pipeline and the description algorithm enable the application to identify and locate various food items clockwise on the plate, and provide precise acoustic descriptions to users. A blind co-designer proposed the concept for the application and participated in two rounds of testing during development to refine it. These tests included comparing the application's precise localization information with the descriptive information provided by Be My Eyes. Our application has the potential to improve the dining experience for blind people by providing an intuitive overview of the placement of food items on a plate, eliminating the need for exploratory efforts. We also discuss the remaining challenges, e.g., capturing high-quality photos of the plate, and potential solutions.

CCS Concepts

• **Software and its engineering;**

Keywords

Visual impairment, iOS application, Semantic segmentation, Food classification

ACM Reference Format:

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1 Introduction

Blind people often have difficulties in identifying and locating food items on their plate when eating. Although they know what they have ordered, they are not aware of the exact locations of the various components of the meal on a plate. The problem can be partially solved by touching the food, but there are still some challenges [26]: apart from hygiene issues, blind people could be pricked by sharp bones or hurt by hot food. Moreover, tactile exploration is inconvenient when dealing with liquids. Using cutlery (e.g., knives, and forks) to detect food can also be a challenge, as they only give information regarding the food's texture (strong or low resistance) and shape.

There are assistive applications designed to help blind people in this case. Be My Eyes [13] provides immediate visual aid assistance by connecting users with a global network with descriptions powered by ChatGPT [24]. When eating, users can capture a photo without precise positioning and receive a vivid food description generated by ChatGPT. However, this lacks sufficient granularity to provide complete detailed information for users. Blind people often use clock directions to describe and locate objects in their vicinity [2, 31]. For example, *12 o'clock* means directly in front, *3 o'clock* means to the right, *6 o'clock* means behind. This method helps them to understand the surroundings and describe the position of objects more accurately. For food descriptions of a plate, the center of the plate is treated as the center of the clock, with *6 o'clock*

positioned in the direction towards the user. This idea is adopted by our **DishDetect** application. To our knowledge, DishDetect is the first application that combines food segmentation based on Artificial Intelligence (AI) with clockwise descriptions to describe what is on a plate. It is not only a lightweight food identification application deployed on iOS, but also an important extension of existing visual aids for blind people.

Segment Anything(SAM) [15] is a foundation model for image segmentation. The SAM-based FoodSAM [16] is an open source model that achieves state-of-the-art performance in food segmentation in a wide range of food categories. DishDetect utilizes FoodSAM to obtain the segmentation result. Subsequently, we developed an accessible iOS application featuring a user-friendly interface that enables users to take a photo of a plate and receive audio descriptions of the food items, organized by clock zones. We host the large segmentation model on a local server, while the iOS application handles only the interaction features, which ensures an efficient user experience.

The design and development of DishDetect follows a co-design procedure. The concept for DishDetect was proposed by a blind co-designer, who serves as both a target user and a contributing coauthor of this work. The project was initiated in the university canteen by the co-designer in collaboration with four technical staff members and an accessibility expert. The co-designer also participated in two rounds of application evaluation, providing feedback that guided the refinement of the application. Based on feedback from the co-designer and another blind volunteer, DishDetect performs better than Be My Eyes in terms of positioning accuracy. Finally, we discuss the unresolved challenges, such as the difficulty of capturing a photo focused on the center of the plate, as well as potential alternative solutions. We also outline our plan to recruit more participants to further explore and evaluate our application.

The contribution of this work is that we design an application that clockwise describes the location of food on the plate, proposing the system architecture and the description algorithm. Importantly, we involved a blind co-designer in the whole process and did two rounds of evaluations for refinement, making DishDetect useful for the community.

2 Related Work

In this section, we present an overview of existing applications designed to provide descriptions to blind individuals. We also discuss the task of semantic segmentation and interaction features, which form the technological background of our study.

2.1 Applications for Descriptions

We introduce here various applications designed for scene and image description to assist blind people. ChatGPT [24] can provide comprehensive and vivid descriptions of a photo, which represents a significant advancement for the lives of blind people. Be My Eyes [13], originally an application that connected users with volunteers for real-time assistance, now incorporates ChatGPT to interpret photos and enable further querying, enhancing its functionality and accessibility. ImageExplorer [17] and Seeing AI [21] provide brief captions to describe photos and allow users to explore each part of the photo interactively on a screen. Seeing AI also

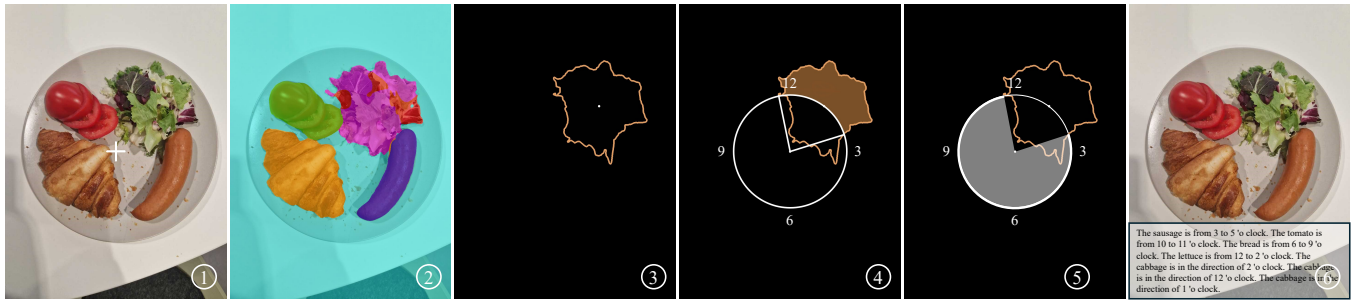


Figure 2: Clock range description algorithm. (1) Input photo taken by a sighted individual. (2) Panoptic segmentation result. (3) Contour and center of the lettuce segment. (4) Part of the lettuce considered for range calculation. (5) Continuous compensate range of the largest blank range. (6) Food description.

includes features for detecting banknotes and scanning barcodes. Envision [11] also describes photos and offers an optional pair of smart glasses to capture an egocentric view. Lookout [12] and Aipoly [30] identify specific objects in the view with the highest confidence scores, while WorldScribe [10] generates a description of the current scene, with its level of detail depending on how long the viewpoint is maintained. The aforementioned applications serve as references for the development of our functions and user interface.

2.2 Semantic Segmentation

Semantic segmentation is a computer vision task that is widely used to make dense predictions in real-world applications such as autonomous driving [8, 25], medical image [9, 34], and satellite image processing [5]. For food segmentation, there are various datasets, such as Foodseg103 [32], UEC FOODPIX COMP [20] and Food-101 [7], which are often used to obtain annotations for food segmentation [22, 36]. Regarding semantic segmentation models, FCN [19] was among the pioneers in using convolutional neural networks (CNN) [5], U-Net [29] with an encoder-decoder architecture designed for biomedical image segmentation. SAM [15] is a cutting-edge foundation model designed to be versatile and capable of handling a wide range of segmentation tasks without the requirement of fine-tuning. Efficient SAMs [35, 38] are proposed to reduce latency while maintaining performance. Liu *et al.* [18] are the first to combine SAM with a captioning model for scene understanding to assist blind people. FoodSAM [16] achieves state-of-the-art identifying extensive food categories effectively by combining SAM with a semantic segmenter. To bypass the need for data set collection, we utilize FoodSAM for food identification in our application.

2.3 Interaction Features

We delve here into the specific user interface and interaction features in existing assistive applications that could help blind users interact with food-related contents. Be My Eyes [13] has a user interface that emphasizes audio-first interaction through a clean, minimalist interface. Seeing AI [21] implements a multimodal interface with distinct swipeable screens for different functions. For object recognition, it employs continuous audio feedback during camera movement and haptic feedback when touching an object

segment in a photo. TapTapSee [28] focuses on a minimalist user interface design with large touch targets. The application uses a simple double tap anywhere on the screen to capture a photo, eliminating the need for a precise location of the button. We summarize that common user interface features for blind and visually impaired people [14] [39] [1] in these applications include simplified visual layouts [4] [27] [13] [21], large touch areas [17] [30], audio feedback for all actions [10] [21], haptic confirmation signals [13] [24], gesture-based controls [17] [21] and voice guidance for camera positioning [12] [13]. Building on this foundation, DishDetect offers a user-friendly interface and interactive features, while also providing structured spatial information about the location of food items.

3 DishDetect

The main idea of creating a system that describes food on a plate clockwise was developed by a blind co-designer. Before the beginning of the application design, our co-designer demonstrated to the developers how mobility trainers describe food and how blind individuals interact with their mobile phones. Then, six of the coauthors, *that is*, two experts in accessibility and technique, the co-designer, and three students, defined the task in the university canteen during lunch time. We gained valuable insight: Although blind individuals often know what they have ordered, they may not be aware of the placement of items on their plate and need to explore it by themselves. To efficiently convey this information, the application should describe the food segments clockwise, similar to the method used by mobility trainers. For the design and development of a food description application for blind users, we divided the implementation into three parts: (1) food identification, which extracts semantic information from the photo, (2) a description algorithm, which delivers accessible information to the blind users and (3) deployment on cell phones. During the development of the application, the co-designer participated in two rounds of evaluation to refine the application. The procedure is described at the end of this section.

3.1 Food Identification

FoodSAM [16] for food identification in our application consists of three parts: SAM [15] for class-agnostic binary mask generation, a semantic segmenter for mask-food category match, and an

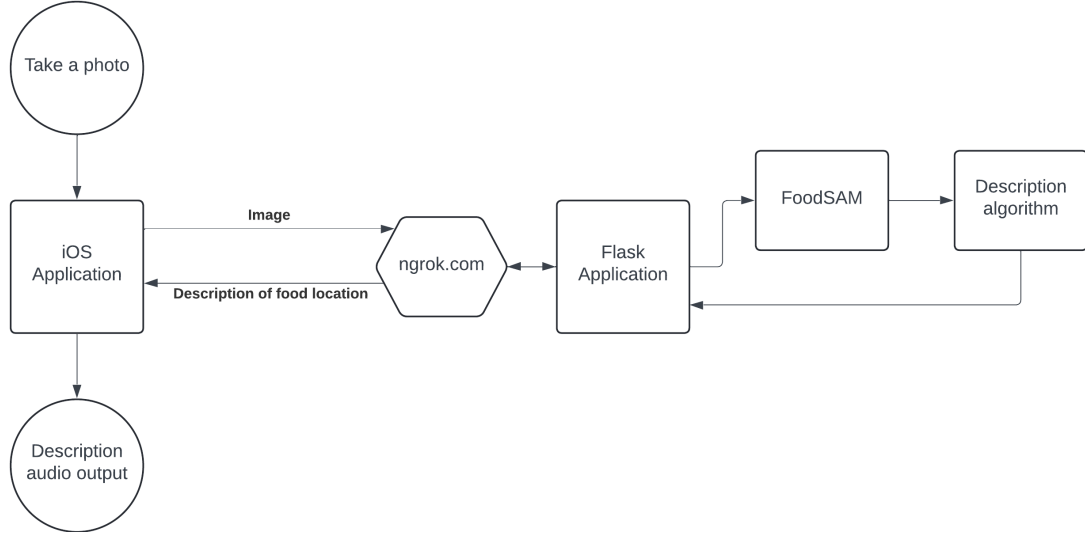


Figure 3: DishDetect software architecture, illustrating the process from image capture to audio output. The system integrates an iOS app, an ngrok tunnel, a Flask application, the FoodSAM model, and a description algorithm.

object detector for non-food categories. The framework enhances the semantic mask with a merge strategy and generates instances and panoramic results. Moreover, FoodSAM extends its zero-shot capability to encompass panoptic segmentation by incorporating an object detector, which enables FoodSAM to effectively capture non-food object information. FoodSAM achieves state-of-the-art performance at 57.44% mean accuracy on FoodSeg103 [33] dataset. Remarkably, this pioneering framework stands as the first work to achieve instance, panoptic, and promptable segmentation on food images.

3.2 Description Algorithm

Here, we propose an algorithm to transform the binary segmentation masks from FoodSAM into accessible information. Each mask in the FoodSAM output corresponds to a food instance (second step in Figure 2). To obtain a complete segment binary mask of a food category, masks with the same category should be merged. For this purpose, the closing morphological dilation is conducted with a kernel size of 5 pixels. Then the unique contour of each merged segment could be found, and the center of the segment would be calculated by using the contour moments (third step in Figure 2). The spatial moments m_{ji} and the central moment mu_{ji} are computed as, where $\bar{x}$ and $\bar{y}$ are the mass center:

$$m_{ji} = \sum_{x,y} \left(\text{array}(x,y) \cdot x^j \cdot y^i \right), \quad (1)$$

$$mu_{ji} = \sum_{x,y} \left(\text{array}(x,y) \cdot (x - \bar{x})^j \cdot (y - \bar{y})^i \right), \quad \bar{x} = \frac{m_{10}}{m_{00}}, \bar{y} = \frac{m_{01}}{m_{00}}. \quad (2)$$

After finding the center, we establish a polar coordinate based on the center of the image to describe the position:

$$r = \sqrt{\bar{x}^2 + \bar{y}^2}, \quad (3)$$

$$\theta = \begin{cases} \arctan\left(\frac{\bar{y}}{\bar{x}}\right), & \bar{x} \neq 0 \\ \pm \frac{\pi}{2}, & \bar{x} = 0. \end{cases} \quad (4)$$

If the distance between the segment and the image center is less than 50 pixels, the food segment will be described as “The <food> is in the center of the plate.” Considering a large piece of food, whose center is more than 50 pixels away from the center of the image, and whose area includes the center of the image, it can be described as “from 10 to 12 o’clock.” with the aforementioned algorithm. Therefore, for the calculation of the clock range, we only consider the part beyond the segment that extends from the center of the segment to the center of the image (the fourth step in Figure 2). The counter of the food segment may intersect with the circle over the center of the segment, so the clock range of the segment can be discontinuous. For this case, we calculate the compensatory clock range of the largest blank area as the food range (the fifth step in Figure 2).

In order to transition from the range of polar angles to a clock-wise range, we reduce the range of polar angles by 15 degrees at both the beginning and the end. This adjustment omits the clock range where only a few pixels of the food appear. Then we align the remaining polar angles θ with the clock C :

$$C = \left\lfloor \frac{((\theta - 15) \bmod 360)}{30} \right\rfloor + 1. \quad (5)$$

The food segments are described with the templates “The <food> is in the direction of <clock> o’clock.” and “The <food> is from <clock_start> to <clock_end> o’clock.”.

3.3 Software Deployment

This section details the software architecture and interaction features of our DishDetect application.

3.3.1 System Architecture. The communication of DishDetect is based on Ngrok [23], which functions as a reverse proxy tool that generates public Uniform Resource Locators (URLs) to expose local servers to the Internet. It is achieved by mapping the local server to a public URL via a secure tunnel, enabling users to access the server. To ensure secure data transfer, all traffic through the tunnel is encrypted. This feature enables developers to share local applications with external users quickly and effortlessly, bypassing complex network configurations.

DishDetect comprises an iOS application and a Flask application, leveraging the FoodSAM model for accurate food segmentation, as shown in Figure 3. The user takes a photo using the iOS application, which will be sent through Ngrok to the Flask application, enabling secure and reliable communication between the mobile application and a server with GPU GTX 1070. The Flask application processes this image with the FoodSAM model to segment the food items and sends the output mask to the description algorithm. Processing images on a Flask server instead of directly on an iOS device reduces the size of the application and decreases its storage demands. Furthermore, the modular design of the system facilitates easy maintenance and scalability, allowing independent updates or scaling of components without disrupting the overall service.

3.3.2 Interaction Features. Our DishDetect application has been designed with accessibility as a primary focus, ensuring that blind users can navigate and use the app. Referring to [3, 6, 13], the simplified user interface enhances the experience of blind users and improves efficiency while avoiding redundant buttons and actions. Our user interface has been optimized for screen readers, with clear labels and hierarchical navigation. All elements are properly tagged to provide contextual information when accessed through assistive technologies. To indicate that the application is processing an image, we have implemented a gentle, non-intrusive waiting sound. This audio cue plays during the image segmentation process, providing feedback to the user that their request is actively being processed. All interactive elements, including the only button “Take a photo”, are designed to be identifiable and operable. They have large touch areas and are positioned for comfortable reach. Each press of the button is accompanied by an audio confirmation.

3.4 Iterative Improvement of the Application with a Blind Co-designer

We tested the app with the co-designer in two cycles, incorporating his feedback. Initially, we used the beta version, which described the food segments only by their central points, not regions, as shown in Figure 4a. After refining the app based on the co-designer's feedback, the advanced version now describes the regions of the food. The co-designer then retested the updated DishDetect, as shown in Figure 4b. After each test of DishDetect, he also used Be My Eyes to compare the functionalities of the two applications, noting that while DishDetect provides precise descriptions, Be My Eyes offers more descriptive details of the food, but lacks specific location information.

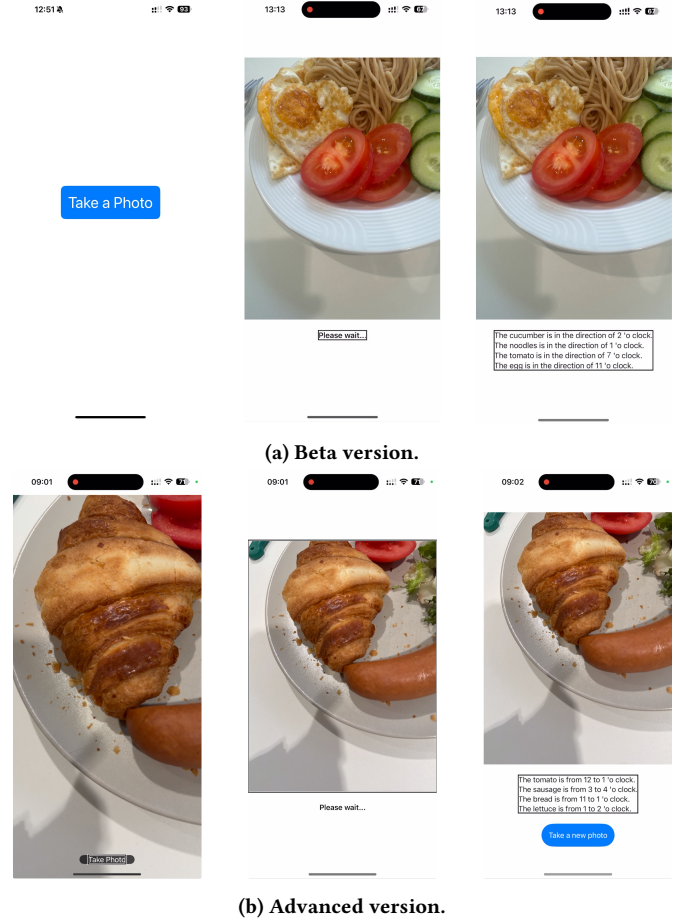


Figure 4: Screenshots taken when the co-designer tested the application: (a) Beta version: the food segments are described by their center points. (b) Advanced version: the food segments are described in a clockwise manner by region.

3.4.1 Beta Version. In contrast to Be my Eyes, which outputs detailed and vivid descriptions, such as “thin slices of cucumber” or “thick slices of tomato”, DishDetect only outputs the category and location of the food, e.g. “The cucumber is in the direction of 2 o'clock, the noodles is in the direction of 1 o'clock...”, which was much briefer and clearer.

In terms of accuracy, the descriptions provided by DishDetect were perceived as precise, although there were a few instances of incorrect positional information. Further analysis of the screenshot video revealed that the outputs were incorrect because of the angle at which the photo was taken, and the center of the photo being regarded as the center of the clock instead of the center of the plate. The co-designer mentioned that “you have to learn a bit and become familiar with the system”. Additionally, the co-designer said, “the app lacks the function to describe food in clock zones, like from x o'clock to y o'clock. It just describes the midpoint of the food.”

In general, the co-designer said: “Be My Eyes and other applications for image description, are unable to describe food clockwise [...]”

As an accessibility study, it works very well.”, and rated DishDetect as “*recommended*”. The location of the food is just as important as the description of the food from his point of view. Therefore, it would be good to combine the functionalities of DishDetect and Be My Eyes in future applications.

3.4.2 Refinement of DishDetect. Based on the feedback from the co-designer, we refined the description algorithm by transitioning from center-point descriptions to region-based descriptions, enabling the generation of more comprehensive outputs. Furthermore, we added music during the image processing time to alert users that the application is running.

3.4.3 Evaluation of the Advanced Version. We re-evaluated the advanced version of DishDetect in collaboration with the co-designer and another volunteer. They were tasked with assessing the system based on human evaluation metrics [37], focusing on positioning accuracy, ease of operation (actionability), and language fluency. The assessment was carried out using a Likert scale between a range of 1=very dissatisfied and 5=very satisfied). Then we conducted a structured interview to gain deeper insights into the feedback and perspectives of the evaluators.

Here, we present the averaged quantitative feedback from the evaluators. In terms of positioning accuracy, the advanced version of DishDetect received a score of 3. In comparison, Be My Eyes received a score of 1, as its descriptions of the food on the plate lacked specific location details. For ease of operation, both applications received a score of 4. Similarly, in terms of language fluency, both the advanced version of DishDetect and Be My Eyes achieved a score of 4.

In addition, we collect qualitative feedback from participants. Compared to DishDetect Beta, users have provided positive feedback on the transition from single point to regional descriptions. *“Like from 7 o’clock to 9 o’clock. this kind of improvement [...] Nice if you have a start, for example from 3 o’clock to 5 o’clock or if your meat is on from 3 o’clock to 4 o’clock. Then you know. OK, this piece is very small.”*

However, users suggested that the software could be improved by offering more assistance and providing clearer instructions, particularly with regard to focusing on the center of the plate. The co-designer suggested that *“you have to improve the thing if I do not have the center of the plate in focus [of the camera].”*

4 Discussion and Future Work

During the evaluation, users encountered difficulties while taking photos. The application could be improved by adding a notification, using the gyroscope to indicate if the photo is horizontal, and integrating an object detection algorithm to locate the center of the plate. However, these enhancements are challenging to implement on a smartphone due to the variety of plate shapes and the difficulty of capturing a complete image of the plate. An alternative solution could be to implement the functionalities of DishDetect on the self-checkout machines in university canteens. This approach would ensure that horizontal photos are taken and that the entire plate is included in the photo.

Locally, we plan to collaborate with canteens. Automatic check-out machines, equipped with precise calibration, can capture high-quality photos that include not only the complete plate but also the knives and forks. We aim to develop an application that receives descriptions using cell phones from the checkout machines via an NFC connection. This information will then be conveyed to blind users through their bone-conduction headsets. With this approach, no additional hardware is required, and users do not need to learn how to take high-quality photos.

In addition, although FoodSAM is a state-of-the-art food segmentation model, it is too cumbersome to deploy on mobile devices and limited in the range of food categories it can identify. For future development, to better adapt to diverse culinary cultures and reduce the latency caused by the large model size, we aim to train a compact food segmentation model using a localized food database.

We also plan to make a more comprehensive user study plan and recruit more target users, to know their needs for food description, suggestions for the improvement of the software, and their preferences of brief or vivid descriptions. We expect conference attendees with blindness or working with blind people to experience our application. Besides, the exploratory user study to evaluate our method will also be conducted within our region, thereby eliciting a more substantial array of opinions and enhancing the popularity of our application, ultimately leading to an augmentation in the user base.

5 Conclusion

This work presents the design and development of an application, DishDetect, which describes food on a plate clockwise and provides information adapted to the eating habits of blind users.

DishDetect was developed using a codesign approach, with the concept proposed by the codesigner, who is both a target user and a coauthor. The project was initiated in the university canteen through collaboration between the co-designer and four technical staff members. We designed a system architecture that integrates a state-of-the-art food identification model and proposed a description algorithm to realize the concept. The co-designer participated in two evaluation rounds, providing feedback that improved the application. Feedback from the co-designer and a volunteer suggests that DishDetect outperforms Be My Eyes in positioning accuracy.

In general, our contributions lay the foundation to future systems that support blind people by giving them an overview of what items are on a plate. We hope that DishDetect, with its refined interaction features and the ability to provide food descriptions similar to those offered by mobility trainers, will be a useful tool for the community.

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References

- [1] Nancy Alajarmeh. 2021. Non-visual access to mobile devices: A survey of touch-screen accessibility for users who are visually impaired. *Displays* 70 (2021), 102081.
- [2] Haikel Alhichri, Yakoub Bazi, and Naif Alajlan. 2020. Assisting the Visually Impaired in Multi-object Scene Description Using OWA-Based Fusion of CNN Models. *Arabian Journal for Science and Engineering* 45, 12 (01 Dec 2020), 10511–10527. <https://doi.org/10.1007/s13369-020-04799-7>
- [3] Fernando Alonso, José L. Fuertes, Ángel L. González, and Loïc Martínez. 2008. User-Interface Modelling for Blind Users. In *Computers Helping People with Special Needs*, Klaus Miesenberger, Joachim Klaus, Wolfgang Zagler, and Arthur Karshmer (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 789–796.
- [4] Mars Ballantyne, Archit Jha, Anna Jacobsen, J Scott Hawker, and Yasmine N El-Glaly. 2018. Study of accessibility guidelines of mobile applications. In *Proceedings of the 17th international conference on mobile and ubiquitous multimedia*. 305–315.
- [5] Alexandre Benoit, Badih Ghattas, Emma Amri, Joris Fournel, and Patrick Lambert. 2021. *Deep Learning for Semantic Segmentation*. Springer International Publishing, Cham, 39–72. https://doi.org/10.1007/978-3-030-74478-6_3
- [6] Tigmanshu Bhatnagar, Youngjun Cho, Nicolai Marquardt, and Catherine Holoway. 2019. Expressive haptics for enhanced usability of mobile interfaces in situations of impairments. *arXiv:1904.06119 [cs.HC]*
- [7] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. 2014. Food-101 – Mining Discriminative Components with Random Forests. In *Computer Vision – ECCV 2014*, David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars (Eds.). Springer International Publishing, Cham, 446–461.
- [8] Senay Cakir, Marcel Gauß, Kai Häppeler, Yassine Ounajjar, Fabian Heinle, and Reiner Marchthaler. 2022. Semantic Segmentation for Autonomous Driving: Model Evaluation, Dataset Generation, Perspective Comparison, and Real-Time Capability. <https://doi.org/10.48550/arXiv.2207.12939>
- [9] Hu Cao, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Qi Tian, and Manning Wang. 2023. Swin-Unet: Unet-Like Pure Transformer for Medical Image Segmentation. In *Computer Vision – ECCV 2022 Workshops*, Leonid Karlinsky, Tomer Michaeli, and Ko Nishino (Eds.). Springer Nature Switzerland, Cham, 205–218.
- [10] Ruei-Che Chang, Yuxuan Liu, and Anhong Guo. 2024. WorldScribe: Towards Context-Aware Live Visual Descriptions. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–18.
- [11] Envision. 2024. Envision Glasses. <https://www.letsenvision.com/de/home>. Accessed: 2024-08-26.
- [12] Google. 2024. Lookout. <https://play.google.com/store/apps/details?id=com.google.android.apps.accessibility.reveal>. Accessed: 2024-06-14.
- [13] Parminder Kaur, Mayuri Ganore, Rucha Doiphode, Ashwini Garud, and Tejaswini Ghuge. 2017. Be My Eyes : Android App for visually impaired people. <https://doi.org/10.13140/RG.2.2.12307.48164>
- [14] Akif Khan and Shah Khuro. 2022. A mechanism for blind-friendly user interface adaptation of mobile apps: a case study for improving the user experience of the blind people. *Journal of Ambient Intelligence and Humanized Computing* 13 (05 2022). <https://doi.org/10.1007/s12652-021-03393-5>
- [15] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross B. Girshick. 2023. Segment Anything. 2023 *IEEE/CVF International Conference on Computer Vision (ICCV)* (2023), 3992–4003. <https://api.semanticscholar.org/CorpusID:257952310>
- [16] Xing Lan, Jiayi Lyu, Hanyu Jiang, Kun Dong, Zehai Niu, Yi Zhang, and Jian Xue. 2023. FoodSAM: Any Food Segmentation. *IEEE Transactions on Multimedia* (2023), 1–14. <https://doi.org/10.1109/TMM.2023.3330047>
- [17] Jaewook Lee, Jaylin Herskovitz, Yi-Hao Peng, and Anhong Guo. 2022. Image-Explorer: Multi-Layered Touch Exploration to Encourage Skepticism Towards Imperfect AI-Generated Image Captions. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). ACM, New York, NY, USA, Article 462, 15 pages. <https://doi.org/10.1145/3491102.3501966>
- [18] Ruiping Liu, Jiaming Zhang, Kunyu Peng, Junwei Zheng, Ke Cao, Yufan Chen, Kailun Yang, and Rainer Stiefelhagen. 2023. Open Scene Understanding: Grounded Situation Recognition Meets Segment Anything for Helping People with Visual Impairments. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*. 1857–1867.
- [19] Jonathan Long, Evan Shelhamer, and Trevor Darrell. 2015. Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 3431–3440.
- [20] Yoshiyuki Matsuda, Hiroki Hoashi, and Keiji Yanai. 2012. Recognition of multiple-food images by detecting candidate regions. In *2012 IEEE International Conference on Multimedia and Expo*. IEEE, 25–30.
- [21] Microsoft. 2017. Seeing AI: Turning the visual world into an audible experience. <https://www.microsoft.com/en-us/seeing-ai/>
- [22] Wei Min, Jie Jiang, and Chongyang Liu. 2019. FoodSegNet: A Semantic Segmentation Framework for Food Images. *IEEE Transactions on Multimedia* 21, 12 (2019), 3199–3211.
- [23] ngrok. 2023. ngrok Documentation. <https://ngrok.com/docs> Accessed: 2023-10-14.
- [24] OpenAI. 2024. ChatGPT: A Large Language Model. <https://openai.com/chat>. Accessed: 2024-10-19.
- [25] Ilias Papadeas, Lazaros Tsochatzidis, Angelos Amanatiadis, and Ioannis Pratikakis. 2021. Real-Time Semantic Image Segmentation with Deep Learning for Autonomous Driving: A Survey. *Applied Sciences* 11, 19 (2021). <https://doi.org/10.3390/app11198802>
- [26] Boris Schauerte, Manel Martinez, Angela Constantinescu, and Rainer Stiefelhagen. 2012. An Assistive Vision System for the Blind That Helps Find Lost Things. In *Computers Helping People with Special Needs*, Klaus Miesenberger, Arthur Karshmer, Petr Penaz, and Wolfgang Zagler (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 566–572.
- [27] Javier Sánchez Sierra and J Togores. 2012. Designing mobile apps for visually impaired and blind users. In *The Fifth international conference on advances in computer-human interactions*. Citeseer, 47–52.
- [28] TapTapSee Team. [n. d.]. TapTapSee: Assistive Technology for the Blind and Visually Impaired. <https://taptapseeapp.com/>
- [29] Hans Thisanke, Chamli Deshan, Kavindu Chamith, Sachith Seneviratne, Rajith Vidanaarachchi, and Damayanthi Herath. 2023. Semantic Segmentation using Vision Transformers: A survey. <https://doi.org/10.48550/arXiv.2305.03273>
- [30] V7. 2024. Aipoly. <https://www.aipoly.com>. Accessed: 2024-07-04.
- [31] Mohammad Moeen Valipour and Anglica de Antonio. 2023. Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping. *Universal Access in the Information Society* 22, 3 (01 Aug 2023), 983–1005. <https://doi.org/10.1007/s10209-022-00868-w>
- [32] Xiongwei Wu, Xin Fu, Ying Liu, Ee-Peng Lim, Steven C.H. Hoi, and Qianru Sun. 2021. A Large-Scale Benchmark for Food Image Segmentation. In *Proceedings of the 29th ACM International Conference on Multimedia (Virtual Event, China) (MM '21)*. Association for Computing Machinery, New York, NY, USA, 506–515. <https://doi.org/10.1145/3474085.3475201>
- [33] Xiao Wu, Peng Xu, Tianyi Lin, and Jiajun Luo. 2021. FoodSeg103: A Challenging Food Segmentation Dataset. *arXiv preprint arXiv:2105.10933* (2021).
- [34] Ruixin Yang and Yingyan Yu. 2021. Artificial Convolutional Neural Network in Object Detection and Semantic Segmentation for Medical Imaging Analysis. *Frontiers in Oncology* 11 (03 2021), 638182. <https://doi.org/10.3389/fonc.2021.638182>
- [35] Chaoning Zhang, Dongshen Han, Yu Qiao, Jung Uk Kim, Sung-Ho Bae, Seungkyu Lee, and Choong Seon Hong. 2023. Faster Segment Anything: Towards Lightweight SAM for Mobile Applications. *arXiv preprint arXiv:2306.14289* (2023).
- [36] Wei Zhang and Yi Liu. 2020. Adapting Food-101 for Image Segmentation via Supervised Learning and Annotation Tools. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*.
- [37] Yi Zhao, Yilin Zhang, Rong Xiang, Jing Li, and Hillming Li. 2024. Vialm: A survey and benchmark of visually impaired assistance with large models. *arXiv preprint arXiv:2402.01735* (2024).
- [38] Chong Zhou, Xiangtai Li, Chen Change Loy, and Bo Dai. 2023. EdgeSAM: Prompt-In-the-Loop Distillation for On-Device Deployment of SAM. *arXiv preprint arXiv:2312.06660* (2023).
- [39] Tiejun Zhu and Yujin Yang. 2024. Research on immersive interaction design based on visual and tactile feature analysis of visually impaired children. *Heliyon* 10, 1 (2024), e22996. <https://doi.org/10.1016/j.heliyon.2023.e22996>